

## SE LAB 3

SANAPALA VENKATA MANEESH

PES1UG24AM247

### Interfaces:

| #  | Interface Connection       | Purpose                                                                                                                                        |
|----|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| I1 | Authentication / Pseudo-ID | Employee Interface → Anonymous Identity Service. Generates a pseudo-ID for the employee.                                                       |
| I2 | Identity Lookup            | Anonymous Identity Service → Appointment & Booking Service. Provides the anonymized identity required for confidential booking.                |
| I3 | Assessment Data            | Employee Interface → Wellness Assessment Service. Sends the employee's stress assessment information.                                          |
| I4 | Wellness Data              | Wellness Assessment Service → Wellness Data Service. Stores assessment results for future tracking.                                            |
| I5 | Appointment Information    | Employee Interface → Appointment & Booking Service. Sends the employee's counseling appointment request.                                       |
| I6 | Session Information        | Therapist Interface → Therapist Session Management Service. Allows the therapist to access and manage counseling sessions using the pseudo-ID. |
| I7 | Wellness History           | Therapist Session Management Service → Wellness Data Service. Retrieves relevant wellness information when required.                           |
| I8 | Pseudo-ID Validation       | Therapist Session Management Service → Anonymous Identity Service. Validates the employee pseudo-ID without exposing corporate identity.       |

### Architecture Analysis:

| Factor                           | Layered Architecture                                                    | Microservices Architecture                                                              | Client–Server Architecture                                                |
|----------------------------------|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| How it would work for our system | We could separate the employee and therapist interfaces, business logic | We could have separate services for identity, wellness assessment, appointment booking, | The employee and therapist would communicate with one central server that |

| Factor                              | Layered Architecture                                                                                                                       | Microservices Architecture                                                                                                 | Client–Server Architecture                                                                                   |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
|                                     | and wellness data into different layers.                                                                                                   | therapist sessions and wellness data.                                                                                      | handles the main functions of the system.                                                                    |
| <b>What works well</b>              | It is simple to understand and keeps different responsibilities separated.                                                                 | Different functions can be handled separately and each service can be scaled when needed.                                  | It is simple to deploy and keeps the system's data and control centralized.                                  |
| <b>Problem for our case</b>         | The different functions may become more dependent on common layers. It is also harder to scale only one function when its usage increases. | It is more complicated to manage because several services have to communicate with each other.                             | A central server could become a bottleneck. If the server fails, many parts of the system could be affected. |
| <b>Employee identity protection</b> | Moderate — identity protection can be implemented, but it may be spread across different layers.                                           | Strong — a separate Anonymous Identity Service can handle pseudo-IDs and keep corporate identity separate.                 | Moderate — identity information would be handled by the central server.                                      |
| <b>Scalability</b>                  | Moderate — scaling individual functions is more difficult.                                                                                 | Strong — services such as booking or assessment can be scaled independently.                                               | Moderate — the central server may become overloaded as usage increases.                                      |
| <b>Performance</b>                  | Moderate — requests may have to pass through multiple layers.                                                                              | Strong — individual services can process and scale independently, although communication between services must be managed. | Moderate — a heavily loaded central server could affect response time.                                       |
| <b>Overall for our case</b>         | Good, but not the best fit                                                                                                                 | <b>Best fit</b>                                                                                                            | Good, but has scalability concerns                                                                           |

## Final Choice For Architecture: Microservices Architecture

### Justification of the Choice:

- We selected **Microservices Architecture** because our Corporate Mental Wellness & Counseling Platform has several different functions such as employee wellness assessment, anonymous identity

management, counseling appointment booking, therapist session management and wellness tracking.

- It helps us protect employee identity. The **Anonymous Identity Service** can generate a pseudo-ID for the employee so that the therapist does not see the employee's corporate email or employee ID.
- Each major function can have its own service. For example, wellness assessment, appointment booking and therapist session management can be separated instead of putting all the processing into one large application.
- It supports independent scaling. If the number of employees using the assessment or appointment service increases, that particular service can be scaled without scaling the complete system.
- It helps with system maintenance. If we need to modify the appointment booking service, we can work on that service without changing the entire application.
- It provides better fault isolation. If one service, such as the wellness tracking service, has a problem, other services such as assessment or appointment booking can continue working.
- It provides a performance benefit because services handling high-demand functions can be optimized and scaled independently.
- It also gives flexibility for future growth. If the system later needs additional wellness features, new services can be added without redesigning the whole application.
- There are some drawbacks, such as greater complexity, communication between services and possible data consistency issues. However, for our Corporate Mental Wellness & Counseling Platform, the benefits of independent scaling, separation of responsibilities and employee identity protection make **Microservices Architecture** the most suitable choice.

## Component and Interface Diagram:

Component and Interface Diagram

